

ETHAN MOON

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EDUCATION

B.S. Engineering (Incoming) | *University of Michigan, College of Engineering* Aug 2026

High School Diploma (In Progress) | *Forest Hills Central High School* Aug 2022 – Present

- GPA (Weighted): 4.342 | SAT Superscore: 1520 (Math: 770, English: 750)
- 15+ AP Exams: Chemistry, Physics C (Mech & E&M), Calculus BC, CS A, Biology, Spanish, Macro/Microeconomics, and more
- Dual Enrollment: Multivariable Calculus, Differential Equations with Linear Algebra — Calvin University

RESEARCH & ENGINEERING EXPERIENCE

Bioprinting Researcher / Co-Author | *Grand Valley State University (Mentor: Dr. Abishek Kamaraj)* Oct 2025 – Present

- Co-authored & presented at ASEE North Central Conference on a closed-loop bioprinting system integrating HS, undergrad, and grad researchers
- Developed CFD digital twin (OpenFOAM/interFoam) to model bio-ink extrusion; built real-time computer vision + sensor feedback pipeline for defect detection and adaptive control
- Preparing research for bioprinting journal publication

Hysteresis Researcher | *Grand Valley State University* Summer 2025

- Aerodynamic hysteresis: Designed/3D-printed airfoil prototypes, ran wind tunnel experiments with Arduino sensors, analyzed flow via CFD

Satellite Dynamics Researcher | *Grand Valley State University* Summer 2025

- Satellite orbital dynamics: Simulated orbital transfers and relative trajectories using Octave & NASA GMAT; implemented rendezvous and formation-flying algorithms

Internship — Software Simulation Intern | *Casual Simulations* Mar 2024 – Present

- Optimized rowing system simulations using XP exchange and Arduino; sold rowing app to a high school rowing team

WMU Design Build Fly Team | *Western Michigan University (High School Participant)* Jan 2025 – Present

- Collaborated with university engineers to design, build, and test model aircraft for AIAA collegiate competitions

Independent Project — "Healthy Eaters" | *Self-Directed* 2024

- Designed a sensor-based smart plate (Arduino/ESP32 + load cells) + gamified Lua/Defold app to improve dietary habits in children with autism; real-world testing showed increased food variety

LEADERSHIP & ACTIVITIES

Founder / Head Coach | *Helping Hands N.P.O. (helpinghandstraining.com)* Feb 2022 – Present

- Founded nationwide volunteer club delivering soccer education to underserved youth; recruited 10+ ECNL/MLS Next coaches across the U.S.; raised \$3,000+ in donations

Senior Editor | *International STEM Research Journal (Ph.D. mentorship)* Apr 2024 – Present

- Produce editorials and provide copy-editing for biannual STEM publication; mentored as Junior Editor (2024), promoted to Senior Editor (Jan 2025)

Varsity Soccer Captain | *Forest Hills Central HS / MLS NEXT (5 years)* Aug 2020 – Present

- All-Conference, All-District, All-Region 1st Team, All-State Honorable Mention (2024); Vice-Captain MLS NEXT 2023–24

MYLEAD Leadership Camp | *Michigan Youth Leadership (School Representative → Alumni Mentor)* Summer 2024 – 2025

Senior Analyst | *FH Charitable Investments* Aug 2023 – Present

- Lead team facilitating charitable investments and financial education for underserved communities

Advanced Math Instructor | *Mathnasium* Dec 2025 – Present

- Deliver personalized instruction in algebra through precalculus; diagnose learning gaps and design targeted improvement plans

Volunteer — Planetarium & Astrophotography Exhibit Director | *Grand Rapids Public Museum* Apr 2024 – Present

- Initiated and developed astrophotography exhibit; sourced 25+ high-res images donated to museum, now presented as interactive discovery cart

HONORS & AWARDS

Academic:

- DECA State Champion — Integrated Marketing Campaign (Feb 2026); qualified for ICDC
- 2nd Place Overall, Southwestern Michigan Science & Engineering Fair (2025); advanced to Michigan State SEEF
- Dartmouth Book Award (2025) | AP Scholar with Distinction | National Recognition Program Award (Top 5%)
- 2nd Place, GYM International STEM Innovation Challenge (2025) | Finalist, IAAC (2025)

Violin:

- 1st Prize — Golden Classical Music Awards: Tokyo Opera City Hall (2025), Walt Disney Concert Hall LA (2024), Carnegie Hall NY (2024)
- Grand Prize Virtuoso — Oxford International Music Competition; performed at Sheldonian Theatre, University of Oxford (2024)
- 2nd Place — VIVO International Competition (Merkin Concert Hall); Stulberg International String Competition Master Class participant

SKILLS & TECHNOLOGIES

Programming: Python, Java, MATLAB/Octave, Lua, Arduino (C++)

Engineering: CFD (OpenFOAM), NASA GMAT, CAD, 3D Printing, Laser Cutting, TensorFlow, PyTorch, Git, Simulation Frameworks

Software & Tools: Microsoft Office (Excel, PowerPoint, Word), Google Workspace (docs, slides, sheets)

AI / ML: TensorFlow, PyTorch, neural networks, computer vision, anomaly detection

Certifications: IBM AI Engineering (Coursera), MIT Aerospace, MIT Beaver Works Drone Racing, U-M AI for Design

Languages: English (fluent), Korean (fluent), Spanish (conversational)